Pointwise Limits of Differential-Operator Iterates

Working notes — Jake Edmonstone

A continuation of *Iteration of Linear Differential Operators*

These notes are a place to reconstruct the continuation in my own words. The goal is to understand the real scalar case $P = D^2$, building on the original DRP write-up. Statements below are targets to prove; the proof drafts are unfinished.

Contents

1 From the original problem to second derivatives . . . 2
1.1 What I already know . . . 2
1.2 The first new problem . . . 2
1.3 The bounds I need . . . 2
2 Reconstructing the proof for D^2 . . . 3
2.1 Two endpoints control a short interval . . . 3
2.2 The good set and the first Baire argument . . . 3
2.3 Putting the orbit in a sequence space . . . 4
2.4 Eliminating the exceptional set . . . 4
2.5 Passing to the limit . . . 5
3 Questions to bring to a meeting . . . 5

1 From the original problem to second derivatives

1.1 What I already know

For ordinary differentiation, the original project proves that $f^{(n)}(x) \to g(x)$ at every point forces g to be smooth and to satisfy $g' = g$. Pointwise convergence gives pointwise bounds on all derivatives; Tao's theorem gives an entire extension, which supplies uniform bounds on compact intervals. Those bounds justify passing to the limit in the integral identity.

To write. Summarize that argument in my own words. Identify exactly where pointwise convergence alone would be insufficient, and explain how the coordinate change and integrating factor handle $L = aD + b$ when a is nowhere zero.

1.2 The first new problem

Throughout the first proof, let $I \subseteq \mathbb{R}$ be a nonempty open interval, $f \in C^\infty(I)$ be real valued, and $D = \frac{\mathrm{d}}{\mathrm{d}x}$. Write $\mathbb{N}_0 = \{0, 1, 2, \ldots\}$ and define

$$u_n := (D^2)^n f = f^{(2n)}, \quad n \in \mathbb{N}_0.$$

The relation linking consecutive iterates is

$$u_n'' = u_{n+1}.$$

Theorem 1.1 **Second-derivative limit — target statement**

Suppose $u_n(x) \to g(x)$ for every $x \in I$. Then $g \in C^\infty(I)$, $g'' = g$, and $u_n \to g$ in $C^\infty_{\mathrm{loc}}(I)$. In particular,

$$g(x) = Ae^x + Be^{-x}$$

for real constants A, B.

To write. Restate the theorem without notation for local smooth convergence. Explain why knowing the even derivatives are pointwise bounded does not immediately let me use the original all-derivatives theorem.

1.3 The bounds I need

For a compact interval $K \subseteq I$ and $k \geq 0$, put

$$\|h\|_{C^k(K)} := \max_{0 \leq j \leq k} \sup_{x \in K} \left|h^{(j)}(x)\right|.$$

Convergence in $C^\infty_{\mathrm{loc}}(I)$ means that these norms of $u_n - g$ tend to zero for every such K and k.

Pointwise convergence gives

$$M(x) := \sup_{n\geq 0} |u_n(x)| < \infty \quad (x \in I).$$

The main task is to obtain

$$\sup_{n\geq 0} \|u_n\|_{C^k(K)} < \infty$$

for every K and k. First obtain local uniform bounds for the functions; then use ${u_n}'' = u_{n+1}$ to control their derivatives.

2 Reconstructing the proof for D^2

2.1 Two endpoints control a short interval

Lemma 2.1 **Scalar endpoint estimate**

Let $c < e$, $\ell = e - c$, and $h \in C^2([c, e])$. Then

$$\|h\|_{C^0([c,e])} \leq \max\{|h(c)|, |h(e)|\} + \frac{\ell^2}{8}\|h''\|_{C^0([c,e])}.$$

Proof draft. Subtract the affine interpolant

$$a_h(x) := \frac{e - x}{\ell} h(c) + \frac{x - c}{\ell} h(e).$$

The difference $h - a_h$ vanishes at both endpoints and has second derivative h''.

To write. Derive the integral representation for this difference. Compute the integral of the absolute value of its Green kernel and explain the constant $\frac{\ell^2}{8}$.

2.2 The good set and the first Baire argument

Definition 2.2 **Good and exceptional points**

Define

$$G := \{x \in I \mid M \text{ is bounded on a neighborhood of } x\}, \quad F := I \setminus G.$$

A point belongs to G when a single bound controls all iterates throughout some neighborhood of that point.

Proof draft: G is open and dense. For each positive integer m, set

$$S_m := \{x \in I \mid M(x) \leq m\} = \bigcap_{n \geq 0} \{x \in I \mid |u_n(x)| \leq m\}.$$

To write. Prove that each S_m is relatively closed and that these sets cover I. Apply Baire on a compact interval inside an arbitrary open subinterval of I to prove density of G. Explain separately why G is open.

2.3 Putting the orbit in a sequence space

Let $E = \ell^\infty(\mathbb{N}_0; \mathbb{R})$ be the Banach space of bounded real sequences with the supremum norm. Define the left shift by

$$S : E \longrightarrow E$$
$$(\xi_0, \xi_1, \xi_2, \ldots) \longmapsto (\xi_1, \xi_2, \xi_3, \ldots).$$

For $x \in G$, package all iterates into

$$U(x) := (u_0(x), u_1(x), u_2(x), \ldots).$$

The intended equation is $U'' = SU$, with $\|S\| = 1$ and $\|U(x)\| = M(x)$.

To write. Use local uniform bounds on u_n and ${u_n}''$ to bound ${u_n}'$ on compact intervals in G, then repeat to bound every derivative. Justify differentiating U in the supremum norm. Explain how the equation for (U, U') extends its solution to a finite endpoint of a component of G, and how coordinate evaluation identifies the endpoint values.

> **Remark 2.3** **Point to check carefully**
>
> Although $U(x)$ is a bounded sequence at every point of I, its norm-smoothness has only been established on G. Continuing an ODE solution beyond a good component does not yet identify it with the original orbit there.

2.4 Eliminating the exceptional set

Proof draft: F is empty. Suppose otherwise. Apply Baire again, now to the relatively closed set F, to obtain a small interval J meeting F and a bound m for M on $F \cap J$. Choose $a < x_0 < b$ with $x_0 \in F \cap J$ and $a, b \in G \cap J$.

To write. Make this second Baire argument precise. For each component (c, e) of $G \cap (a, b)$, explain why the endpoint values are bounded by $M_0 = \max\{m, M(a), M(b)\}$. Apply Lemma 2.1 to the sequence-valued solution, choose J short enough to absorb the term containing $\frac{\ell^2}{8}$, and obtain a bound on all of $[a, b]$. State the contradiction.

2.5 Passing to the limit

To write. Explain why local uniform derivative bounds and pointwise convergence imply uniform convergence on compact intervals. For $w = u_n - u_m$, use Taylor's formula and a uniform bound on w'' to show that w' tends uniformly to zero on a smaller compact interval. Repeat for higher derivatives, identify their limits, and pass ${u_n}'' = u_{n+1}$ to $g'' = g$. Finish the proof of Theorem 1.1.

3 Questions to bring to a meeting

To write. Record the exact statement or proof step I cannot yet justify, what I tried, and which hypothesis seems to matter.